

Vulnerability Analysis

1. Port Scanning and Service Enumeration

Port Scanning Tool

Tool: Nmap

Target: scanme.nmap.org (45.33.32.156)

Nmap was used to identify open TCP and UDP ports on the target.

The TCP scan identified:

Port	State	Service
22/tcp	Open	SSH
80/tcp	Open	HTTP
9929/tcp	Open	nping-echo
31337/tcp	Open	Elite

The UDP scan also identified UDP services including:

- 68/udp — DHCP
- 123/udp — NTP
- 1900/udp — UPnP

```
(root@kali)-[/home/kali]
# nmap scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:33 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.086s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
25/tcp    filtered  smtp
80/tcp    open      http
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo
31337/tcp open      Elite

Nmap done: 1 IP address (1 host up) scanned in 4.15 seconds
```

```

(root@kali)-[/home/kali]
# nmap -sU scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:34 PDT
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.082s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:
bb2f
Not shown: 991 closed udp ports (port-unreach)
PORT      STATE      SERVICE
68/udp    open|filtered dhcpc
123/udp   open       ntp
135/udp   filtered   msrpc
136/udp   filtered   profile
137/udp   filtered   netbios-ns
138/udp   filtered   netbios-dgm
139/udp   filtered   netbios-ssn
445/udp   filtered   microsoft-ds
1900/udp  open|filtered upnp

Nmap done: 1 IP address (1 host up) scanned in 1155.64 seconds

```

Network Service Enumeration

Nmap service detection (**-sV**) was used to determine the software and versions associated with open ports.

The detection identified:

Port	Service	Version
22/tcp	SSH	OpenSSH 6.6.1p1
80/tcp	HTTP	Apache httpd 2.4.7
9929/tcp	nping-echo	Nping echo

31337/tcp tcpwrapped/Elite Service detected

```
(root@kali)~[/home/kali]
# nmap -sV scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:36 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.087s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:
bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Li
nux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.22 seconds
```

Vulnerability Scan

Tool: Nessus Essentials

Policy: Basic Network Scan

Target: scanme.nmap.org

Status: Completed

The Nessus summary shows multiple findings, including Critical, Low, and Informational results.

Vulnerability 1 — Unsupported Ubuntu 14.04

Severity: Critical

CVSS: 10.0

Affected service: HTTP / port 80

Finding

Nessus identified the target as running **Ubuntu Linux 14.04**, which is no longer supported by the vendor.

Risk

Because the operating system is end-of-life, it no longer receives normal security updates. This increases the possibility that known or future vulnerabilities could remain unpatched.

Potential Impact

An attacker could potentially exploit unpatched vulnerabilities in the operating system or installed software, potentially affecting the confidentiality, integrity, or availability of the system.

Recommended Mitigation

Upgrade the operating system to a currently supported Ubuntu release and maintain regular security updates.

CRITICAL

Canonical Ubuntu Linux SEoL (14.04.x)

>

Plugin Details

Severity: Critical

ID: 201408

Version: 1.2

Type: combined

Family: General

Published: July 3, 2024

Modified: March 26, 2025

Description

According to its version, Canonical Ubuntu Linux is 14.04.x. It is, therefore, no longer maintained by its vendor or provider.

Lack of support implies that no new security patches for the product will be released by the vendor. As a result, it may contain security vulnerabilities.

Solution

Upgrade to a version of Canonical Ubuntu Linux that is currently supported.

See Also

<http://www.nessus.org/u?5afc4199>

Output

OS : Ubuntu Linux 14.04

Security End of Life : April 25, 2019

Time since Security End of Life (Est.) : >= 7 years

To see debug logs, please visit individual host

Port ▲

Hosts

80 / tcp / www

scanme.nmap.org

Risk Information

Risk Factor: Critical

CVSS v3.0 Base Score: 10.0

CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H

CVSS v2.0 Base Score: 10.0

CVSS v2.0 Vector: CVSS2#AV:N/AC:L/Au:N/C:C/I:C/A:C

Vulnerability Information

CPE: cpe:/o:canonical:ubuntu_linux

Unsupported by vendor: true

Vulnerability 2 — SSH Weak Algorithms

Severity: Medium
Plugin: 90317

Affected service: SSH / port 22
CVSS: 4.3

Finding

Nessus identified weak SSH encryption algorithms supported by the remote SSH server.

Risk

Allowing weak cryptographic algorithms can reduce the security of SSH communications and may allow weaker encryption options to be negotiated.

Potential Impact

The use of weak algorithms could reduce the confidentiality and integrity protections provided by SSH communications.

Recommended Mitigation

Configure the SSH server to disable weak or deprecated cryptographic algorithms and allow only strong, currently recommended algorithms.

MEDIUM

SSH Weak Algorithms Supported

Description

Nessus has detected that the remote SSH server is configured to use the Arcfour stream cipher or no cipher at all. RFC 4253 advises against using Arcfour due to an issue with weak keys.

Solution

Contact the vendor or consult product documentation to remove the weak ciphers.

See Also

<https://tools.ietf.org/html/rfc4253#section-6.3>

Output

The following weak server-to-client encryption algorithms are supported :

aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

The following weak client-to-server encryption algorithms are supported :

aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chacha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

To see debug logs, please visit individual host

Port	Hosts
22 / tcp / ssh	scanme.nmap.org

Plugin Details

Severity: Medium

ID: 90317

Version: 1.4

Type: remote

Family: Misc.

Published: April 4, 2016

Modified: January 12, 2026

Risk Information

Risk Factor: Medium

CVSS v2.0 Base Score: 4.3

CVSS v2.0 Vector: CVSS2#AV:N/AC:M/Au:N/C:P/I:N/A/N

Vulnerability 3 — SSH Terrapin Prefix Truncation

Severity: Medium
CVE: CVE-2023-48795
Affected service: SSH / port 22
CVSS v3: 5.9

Finding

Nessus identified the SSH service as potentially vulnerable to the **Terrapin prefix truncation weakness**.

Risk

The vulnerability can affect the integrity of SSH connections under specific conditions involving vulnerable SSH configurations and algorithms.

Potential Impact

A successful attack could potentially weaken the security properties of an SSH connection, depending on the configuration and attack conditions.

Recommended Mitigation

Update OpenSSH to a version containing the appropriate security fixes or apply the vendor-recommended SSH configuration changes that address the vulnerability.

MEDIUM

SSH Terrapin Prefix Truncation Weakness (CVE-2023-48795)

Description

The remote SSH server is vulnerable to a man-in-the-middle prefix truncation weakness known as Terrapin. This can allow a remote, man-in-the-middle attacker to bypass integrity checks and downgrade the connection's security.

Note that this plugin only checks for remote SSH servers that support either ChaCha20-Poly1305 or CBC with Encrypt-then-MAC and do not support the strict key exchange countermeasures. It does not check for vulnerable software versions.

Solution

Contact the vendor for an update with the strict key exchange countermeasures or disable the affected algorithms.

See Also

<https://terrapin-attack.com/>

Output

Supports following ChaCha20-Poly1305 Client to Server algorithm : aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chaCha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

Supports following CBC Client to Server algorithm : aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chaCha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

Supports following Encrypt-then-MAC Client to Server algorithm : hmac-md5-etm@openssh.com,hmac-sha1-etm@openssh.com,hmac-sha1-96-etm@openssh.com,hmac-md5-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-ripemd160-etm@openssh.com,hmac-ripemd160@openssh.com,hmac-sha1-96-etm@openssh.com,hmac-md5-96-etm@openssh.com,hmac-sha1@openssh.com,hmac-md5@openssh.com,umac-128-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-ripemd160,hmac-ripemd160@openssh.com,hmac-sha1-96,hmac-md5-96

Supports following ChaCha20-Poly1305 Server to Client algorithm : aes128-ctr,aes192-ctr,aes256-ctr,arcfour256,arcfour128,aes128-gcm@openssh.com,aes256-gcm@openssh.com,chaCha20-poly1305@openssh.com,aes128-cbc,3des-cbc,blowfish-cbc,cast128-cbc,aes192-cbc,aes256-cbc,arcfour,rijndael-cbc@lysator.liu.se

more...

To see debug logs, please visit individual host

Port

Hosts

22 / tcp / ssh

scanna.rmap.org

Plugin Details

Severity: Medium

ID: 187315

Version: 1.6

Type: remote

Family: Misc

Published: December 27, 2023

Modified: July 10, 2025

VPR Key Drivers

Threat Recency: No recorded events

Threat Intensity: Very Low

Exploit Code Maturity: PoC

Age of Vuln: 730 days +

Product Coverage: Very High

CVSSV3 Impact Score: 3.6

Threat Sources: No recorded events

Risk Information

Vulnerability Priority Rating (VPR): 5.0

Exploit Prediction Scoring System (EPSS): 0.9331

Risk Factor: Medium

CVSS v3.0 Base Score: 5.9

CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:H/PR:N/UI:N/S:U/C:N/H:HA/N

CVSS v3.0 Temporal Vector: CVSS:3.0/E:P/RL:G/RC:C

CVSS v3.0 Temporal Score: 5.3

CVSS v2.0 Base Score: 5.4

CVSS v2.0 Temporal Score: 4.2

Risk Information

Vulnerability Priority Rating (VPR): 5.0

Exploit Prediction Scoring System (EPSS): 0.9331

Risk Factor: Medium

CVSS v3.0 Base Score: 5.9

CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:H/PR:N
/UI:N/S:U/C:N/I:H/A:N

CVSS v3.0 Temporal Vector: CVSS:3.0/E:P
/RL:O/RC:C

CVSS v3.0 Temporal Score: 5.3

CVSS v2.0 Base Score: 5.4

CVSS v2.0 Temporal Score: 4.2

CVSS v2.0 Vector: CVSS2#AV:N/AC:H/Au:N/C:N
/I:C/A:N

CVSS v2.0 Temporal Vector:
CVSS2#E:POC/RL:OF/RC:C

Vulnerability Information

CPE: cpe:/a:openbsd:openssh

Exploit Available: true

Exploit Ease: Exploits are available

Patch Pub Date: December 18, 2023

Vulnerability Pub Date: December 18, 2023

Reference Information

CVE: [CVE-2023-48795](#)

Vulnerability	Severity	Evidence	Potential Impact	Recommended Mitigation
Ubuntu 14.04 End of Life	Critical / CVSS 10.0	Nessus	Unpatched OS vulnerabilities	Upgrade to supported Ubuntu version
SSH Weak Algorithms	Medium / CVSS 4.3	Nessus	Reduced SSH cryptographic protection	Disable weak SSH algorithms
SSH Terrapin (CVE-2023-48795)	Medium / CVSS 5.9	Nessus	Potential SSH integrity/security degradation	Patch OpenSSH/apply vendor mitigation

Nmap identified port 22/tcp as an open SSH service and service enumeration identified OpenSSH 6.6.1p1. Nessus subsequently identified multiple SSH-related vulnerabilities on port 22/tcp. This provided supporting evidence that the vulnerabilities were associated with an identified active service rather than an unrelated closed port.